

Demonstration of Proposed Features

Project Title: Rising Waters

Introduction

The Rising Waters system is demonstrated through a web-based application that predicts flood risk using Machine Learning. The demonstration highlights the complete workflow of the system, from user input to prediction result, showing how the application functions in real time.

System Demonstration

The application begins with a home page that introduces the Flood Prediction System. Users can navigate to the prediction page, where they enter weather-related parameters such as temperature, humidity, cloud cover, annual rainfall, seasonal rainfall, average June rainfall, and subdivision rainfall. After submitting the data, the system processes the inputs using the trained Machine Learning model and displays the prediction result.

Demonstrated Features

The demonstration includes a user-friendly interface for entering weather parameters, automatic preprocessing of input data, Machine Learning-based flood prediction, probability calculation, and result pages showing either **Flood Risk** or **No Flood Risk**. The application also validates user inputs and displays appropriate messages for invalid data.

Project Output

The system successfully predicts the possibility of flood occurrence based on the given input values. It generates quick and accurate results, enabling users to understand the flood risk and take necessary precautions.

Benefits of Demonstration

The demonstration proves that the application is easy to use, provides fast predictions, and integrates Machine Learning with a web application effectively. It also shows the practical implementation of the proposed solution for real-world flood prediction.

Conclusion

The successful demonstration confirms that the Rising Waters Flood Prediction System meets the project objectives. The application provides accurate flood

predictions through a simple web interface and demonstrates the practical use of Machine Learning for disaster preparedness and early flood warning.